Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

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ARTICLE INFO

Keywords:
structural calibration
functional attribution
process supervision
convex optimization
audit prioritization
knowledge-based systems

ABSTRACT

Knowledge-intensive reasoning systems need verification steps that are locally useful and structurally well placed. This paper presents Structurally-Calibrated Functional Attribution (SC-FMA), a claim-bounded methodology for converting coarse utility or proxy-fidelity signals into auditable verification-step weights. SC-FMA optimizes a Structurally-Calibrated Utility (SCU) objective that balances fidelity to the supplied utility anchor, graph-level structural necessity, redundancy reduction, and bottleneck protection. The objective is convex under the stated regularization conditions and yields a unique calibrated weight vector. On a controlled synthetic step-importance benchmark, SC-FMA QP improves mean Spearman correlation with proxy step labels from 0.483 for raw CIU to 0.608. On PRM800K process-supervision annotations (4,417 samples and 34,219 labeled steps under a frozen hash split), the primary positive real-data signal is the stored `w_struct` ranking (Spearman 0.611); SC-FMA Ridge closely preserves it (0.604), while QP and Projection are downgraded on this route. A fixed-budget PRM800K audit-prioritization readout provides moderate, preliminary real-data support for PRM800K-like audit triage only. The package does not claim downstream PRM training gains, GSM8K/HotpotQA replay validation, production KBS deployment, or formal causal identification.

1. Introduction

Knowledge-intensive reasoning systems increasingly expose intermediate verification steps: a rule fires, an entity path is checked, a retrieval result is inspected, or a reflective language-agent step critiques a partial answer. Process supervision turns such steps into supervision targets, but a scalar local utility signal is not enough. A step may correlate with a correct final answer while being redundant because neighboring rules or reflections supply the same support. Conversely, a rare bottleneck check may receive a weak local signal but still gate downstream inference.

This paper studies this problem as *intervention-based functional attribution for reflective cognition dynamics*. The goal is not to identify true causal estimands. Instead, SC-FMA asks how a supplied utility or proxy-fidelity input should be calibrated when observable trace structure indicates redundancy, structural necessity, or bottleneck status. This framing is deliberately close to knowledge-based systems: it treats verification chains as auditable objects whose local and structural roles can disagree.

The paper makes four contributions. First, it defines SC-FMA, a compact calibration method that converts utility or proxy-fidelity inputs into structurally calibrated verification-step weights. Second, it introduces the SCU objective, which combines fidelity, graph necessity, redundancy, and bottleneck terms in a constrained convex program. Third, it evaluates the method on a controlled synthetic benchmark for step-importance ranking. Fourth, it reports bounded real-data evidence on PRM800K process annotations and an offline audit-prioritization readout. All claims are bounded to methodology and audit prioritization; downstream PRM training, task-specific replay validation, production KBS deployment, and causal identification remain outside the evidence surface.

2. Related Work and Boundary

Process reward models and process supervision assign value estimates to intermediate reasoning steps, often improving verifier-guided search or providing denser feedback than outcome-only labels [4, 8, 16]. Automated process-labeling routes such as Math-Shepherd reduce annotation cost but still rely on step-level signals that can drift under

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distribution shift [17, 20]. SC-FMA is complementary: it does not train a new PRM and does not claim downstream PRM improvement. It calibrates existing step-level utility or proxy-fidelity inputs against trace structure.

Attribution and explanation methods such as Integrated Gradients, SHAP, and rationale evaluation explain which input parts matter for a prediction, but they usually treat components as independently attributable units [5, 10, 15]. Graph-based reasoning frameworks and network robustness studies emphasize that structural roles can be non-local, redundant, or bottleneck-like [1, 11, 19]. SC-FMA brings this structural view into step weighting: it uses graph diagnostics as constraints on supervision weights rather than as a separate visualization layer.

The KBS connection is methodological. Classical expert systems and cognitive architectures make intermediate rule chains inspectable [2, 3, 7, 9]. Modern knowledge-enhanced language systems, knowledge graphs, and ontology-aware pipelines likewise need auditable intermediate evidence [6, 12, 18]. SC-FMA does not build an ontology, certify formal reasoning, or validate a production workflow. It supplies a reproducible weighting layer for already recorded reasoning traces and records this package boundary explicitly as `validated_kbs_workflow=false`.

3. Methodology

For a trace with k reflective or verification steps, let $\tilde{\mathbf{c}} \in \mathbb{R}^k$ denote the normalized utility or proxy-fidelity input. In the binary-correctness CIU setting used by the diagnostic pipeline, this input is a trace-level coarse utility anchor: it records how a trace-level outcome changes under a documented intervention and does not prove that each step has an independent outcome-grounded CIU. Let $\tilde{\mathbf{n}}$ denote normalized structural necessity from graph perturbation diagnostics, $\mathbf{R}$ a redundancy matrix, and $\mathbf{b}$ a bottleneck indicator.

SC-FMA computes weights $\mathbf{w}$ on the probability simplex

$$\mathcal{W} = \{\mathbf{w} \in \mathbb{R}^k_+ : \sum_i w_i = 1\}$$

by minimizing the SCU objective:

$$L(\mathbf{w}) = \alpha\|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta\|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2 + \gamma\, \mathbf{w}^\top \mathbf{R}\mathbf{w} - \delta \sum_i b_i \log(w_i). \tag{1}$$

The four terms encode fidelity to the supplied input, structural alignment, redundancy reduction, and non-zero bottleneck protection. Detailed notation, construction rules, algorithms, and proofs are provided in the supplementary material.

Theorem 3.1 (SCU well-posedness)**.** *If $\alpha + \beta > 0$ and the redundancy matrix is positive semidefinite, or if $\delta > 0$ for at least one bottleneck node, then the SCU objective in Eq. 1 is convex on the simplex. With positive quadratic regularization on the feasible subspace or an active log-barrier term, the minimizer is unique. For non-redundant step pairs, the calibrated weights preserve the joint ordering induced by $\tilde{\mathbf{c}}$ and $\tilde{\mathbf{n}}$ under the stated monotonicity condition; bottleneck nodes with $b_i = 1$ receive strictly positive weight.*

We implement three variants. SC-FMA Ridge uses a closed-form softmax over a linear combination of $\tilde{\mathbf{c}}$ and $\tilde{\mathbf{n}}$. SC-FMA QP solves the full constrained program with redundancy and bottleneck terms. SC-FMA Projection applies a fast topology-constrained simplex projection. The QP variant is the most expressive and is used for the controlled synthetic benchmark; Ridge is the closest real-data approximation on PRM800K.

4. Evaluation

4.1. Data and Baselines

The synthetic calibration benchmark contains 200 traces and 1,027 reflective steps generated with fixed seed 42. It is used for method development, calibration, ablation, and error analysis. Controlled proxy step labels combine local correctness, lexical diversity, position, and graph-derived necessity. The positive real-data route uses PRM800K process-supervision annotations under the frozen v3.6/v3.8 hash split: 4,417 samples and 34,219 labeled steps. GSM8K/HotpotQA task-specific replay and downstream filtering routes were not completed and are not counted as positive validation evidence.

Baselines include raw CIU, gradient attribution, attention rollout, Monte Carlo Shapley, token surprisal, step entropy, span length, relative position, and random ranking. The primary metric is Spearman rank correlation; Kendall τ, NDCG@3, NDCG@5, and fixed-budget audit-prioritization metrics are reported where relevant. Hyperparameter grids, additional ablations, error modes, and efficiency measurements are moved to the supplementary material.

Table 1
Step importance ranking on the controlled synthetic benchmark. Mean metrics are over 200 traces and 1,027 steps. Bootstrap confidence intervals and extended ablations are reported in the supplementary material.

Method	Spearman ρ	Kendall τ	NDCG@3	NDCG@5
SC-FMA QP	**0.608**	0.533	0.921	0.958
SC-FMA Ridge	0.533	0.447	0.889	0.927
MC Shapley (500 perm.)	0.524	0.451	0.895	0.933
Gradient×Input	0.491	0.419	0.881	0.924
Raw CIU	0.483	0.412	0.874	0.919
Attention Rollout	0.461	0.388	0.863	0.912
SC-FMA Projection	0.338	0.278	0.806	0.872
Token Surprisal	0.310	0.252	0.791	0.863
Step Entropy	0.287	0.234	0.782	0.857
Span Length	−0.006	−0.013	0.740	0.826
Relative Position	−0.005	−0.014	0.740	0.825
Random	−0.010	−0.008	0.736	0.825

Table 2
Claim-bounded PRM800K real-data evidence and offline audit-prioritization readout. Metrics use the same frozen split of 4,417 samples and 34,219 labeled steps.

Signal	Spearman ρ	NDCG@25%	Allowed interpretation
`w_struct`	**0.611**	**0.9506**	Primary PRM800K step-ranking signal.
SC-FMA Ridge	0.604	0.9460	Closest SC-FMA approximation.
SC-FMA QP	0.442	0.9439	Audit context only on this route.
Frozen PRM prefix score	0.252	0.8465	In-distribution baseline context only.
Raw local utility	−0.077	0.7221	Direct ablation.
Random	0.006	0.7757	Seeded sanity control.

4.2. Synthetic Step-Importance Ranking

Table 1 shows that SC-FMA QP achieves the strongest synthetic ranking result, improving Spearman ρ by 0.125 over raw CIU. The result supports the methodological claim that structural calibration can correct local-utility weights when the benchmark provides controlled proxy step labels. It does not imply that the QP variant is universally best on naturally occurring data.

4.3. PRM800K Evidence and Audit Readout

The locked PRM800K result is route-specific. The stored `w_struct` ranking is the primary positive signal, with Spearman $\rho = 0.611$ against PRM800K step ratings. Raw local utility is negative on this route ($\rho = -0.077$), and the overlap-limited frozen PRM prefix-score baseline is lower ($\rho = 0.252$). SC-FMA Ridge closely preserves `w_struct` ($\rho = 0.604$), while QP ($\rho = 0.442$) and Projection ($\rho = -0.135$) are downgraded.

The fixed-budget readout in Table 2 supports a bounded operational statement: on PRM800K-like process annotations, `w_struct` and Ridge concentrate high-rated steps better than raw local utility and simple controls. This is moderate, preliminary real-data support for audit prioritization. It is not downstream PRM training, not filtering superiority, not GSM8K/HotpotQA replay validation, and not production KBS deployment.

5. KBS Implications and Limitations

SC-FMA is useful for KBS methodology because it separates four audit questions that are often conflated: whether a step is locally useful, whether it is structurally necessary, whether it duplicates another step, and whether it gates downstream reasoning. In rule-chain, retrieval-augmented, or KG-assisted workflows, these questions map naturally to curation priorities: high-necessity bottleneck checks deserve review, while highly redundant checks may be consolidation candidates. The mapping is a methodological analogy, not a validated system integration.

The main limitations are deliberate. First, the synthetic benchmark is controlled and proxy-labeled, so it supports calibration behavior rather than broad external validity. Second, the PRM800K route supports real step-label ranking and audit prioritization only within a PRM800K-like annotation distribution. Third, current graph construction uses temporal and topical trace structure rather than formal ontology semantics. Fourth, the binary-correctness CIU signal remains a trace-level coarse utility anchor and does not provide independent per-step outcome-grounded CIU. Fifth, the framework does not recover Rubin-style effect claims or Pearl-style do-calculus semantics [13, 14]. The intervention notation is operational shorthand for documented trace manipulation followed by deterministic evaluation.

6. Conclusion

This paper presented SC-FMA, a structurally calibrated method for turning utility or proxy-fidelity inputs into auditable verification-step weights. The SCU objective combines fidelity, structural necessity, redundancy, and bottleneck protection in a constrained optimization problem. Controlled synthetic evidence shows that QP calibration improves step-importance ranking over raw CIU and common baselines. Locked PRM800K evidence gives a narrower result: `w_struct` is the primary real-data signal, and SC-FMA Ridge closely preserves it for audit prioritization. The resulting package is a claim-bounded KBS methodology and audit-prioritization contribution, with downstream PRM training, task-specific replay validation, production KBS deployment, and causal identification left for future work.

Acknowledgments

The authors thank the College of Management Science and Engineering at Beijing Information Science and Technology University for research support.

Funding

This work was supported by the National Social Science Fund of China Project (24BSH018) and the Beijing Natural Science Foundation Project (L252145).

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this work, the authors used OpenAI Codex for language editing, LaTeX packaging assistance, and compliance checks. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability

PRM800K is publicly available from its original source. Derived locked-split reports, audit-prioritization artifacts, and reproduction scripts will be made available by the authors on request. The synthetic benchmark is fully reproducible from `scripts/generate_synthetic_traces.py` with seed 42. Frozen experiment configurations are archived in the accompanying repository.

CRediT authorship contribution statement

Haoran Ma: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. **Ningning Wang**: Methodology, Validation, Writing – Review & Editing, Supervision, Project administration, Funding acquisition.

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